

范围查询(Range)

Description

Let S be a set of n integral points on the x -axis. For each given interval $[a, b]$, you are asked to count the points lying inside.

Input

The first line contains two integers: n (size of S) and m (the number of queries).

The second line enumerates all the n points in S .

Each of the following m lines consists of two integers a and b and defines an query interval $[a, b]$.

Output

The number of points in S lying inside each of the m query intervals.

Example

Input

```
5 2
1 3 7 9 11
4 6
7 12
```

Output

```
0
3
```

Restrictions

$0 \leq n, m \leq 5 \cdot 10^5$

For each query interval $[a, b]$, it is guaranteed that $a \leq b$.

Points in S are distinct from each other.

Coordinates of each point as well as the query interval boundaries a and b are non-negative integers not greater than 10^7 .

Time: 2 sec

Memory: 256 MB

描述

数轴上有 n 个点，对于任一闭区间 $[a, b]$ ，试计算落在其内的点数。

输入

第一行包括两个整数：点的总数 n ，查询的次数 m 。

第二行包含 n 个数，为各个点的坐标。

以下 m 行，各包含两个整数：查询区间的左、右边界 a 和 b 。

输出

对每次查询，输出落在闭区间 $[a, b]$ 内点的个数。

样例

见英文题面

限制

$0 \leq n, m \leq 5 \times 10^5$

对于每次查询的区间 $[a, b]$ ，都有 $a \leq b$

各点的坐标互异

各点的坐标、查询区间的边界 a 、 b ，均为不超过 10^7 的非负整数

时间：2 sec

内存：256 MB